

A finite element method based discretisation of the Hamiltonian operator on triangular meshes*

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Abstract

Differential operators and their spectra have many interesting properties and have found applications in areas like Computer Graphics and Geometry processing. In this paper, we study the discretisation of the Hamiltonian operator, a combination of the Laplace-Beltrami operator (LBO) and a potential function-based diagonal operator. We propose an FEM based discretisation of the Hamiltonian operator, and prove that it satisfies properties like symmetricity, and positive semi-definiteness.

We also compare our discretisation with two prevalent discretisations wherein the diagonal potential operator is approximated as a diagonal matrix. While one does not yield a symmetric matrix, and thus may yield non-orthogonal eigenvectors, the other is not as accurate as ours. We analyse the three methods in terms of orthogonality of eigenvectors, errors in eigenfunctions, and solution of the Hamiltonian version of the Poisson equation, and show experiments to demonstrate higher accuracy of the proposed method.

Keywords: Hamiltonian Operator, FEM based Discretisation, Shape Analysis.

1 Introduction

Geometry Processing deals with the analysis and processing of geometric objects, typically 3D objects represented as surface meshes. Shape deformation (animation), computing correspondence between 3D objects, shape matching, interpolation, data compression are some typical problems in this area. A major source of intrinsic and extrinsic features for 3D objects, represented by their surface or volume in $\mathbb{R}^3$, are differential operators [15] and their spectra. A ubiquitous example is the Laplace-Beltrami operator (LBO), a positive-semidefinite self-adjoint operator, giving an orthonormal basis consisting of its eigenfunctions/eigenvectors, which are often referred to as manifold/mesh harmonics and play a role analogous to Fourier series for functions on manifolds.

Over the past decade, several attempts have also been made to capture features or information that is not intrinsic to the surface, such as the mean curvature [9]. Spectrum of operators like the Hamiltonian operator [4, 5, 1], Dirac operator [10] and Dirichlet-to-Neumann operator [14] have been used to perform extrinsic shape analysis. The Hamiltonian operator has been used for various applications - partial shape matching, constrained/compressed manifold modes,

Hamiltonian Functional maps for correspondence [13], to name a few.

For computational purposes all these operators need to be discretised so that they may be applied to discrete representations of 3D objects, for example, point clouds, triangular and polygonal surface meshes, voxels. The Cotan Laplacian [12] is one of the most popular discretisations of the LBO for triangular meshes. Other equivalent discretisations have been obtained using Finite Element Methods(FEM) and Discrete Exterior Calculus(DEC) [6]. For a survey, refer to [3]. In this work, we will work only with triangular meshes.

Since the Hamiltonian operator differs from the LBO by a scalar potential function, existing discretisations of the Hamiltonian for triangular meshes are obtained by adding the product of the mass/area matrix or its approximation with a diagonal matrix containing values of the potential function at the vertices to the Cotan laplacian [4, 5, 13, 1]. For ease of reference, we will call the existing discretisation that uses the mass matrix with a diagonal potential matrix [1], the *Diag* method, while the other that uses a diagonal approximation to the mass matrix (lumped mass matrix) together with the diagonal potential matrix, the *Lumped* method. For details, refer to Section 2. In this paper, we analyse both the existing discretisations, and provide a better discretisation based on FEM using piece-wise linear functions.

We prove that the proposed method has nice theoretical properties. An example of these properties is shown in Figure 1, where a step potential function with values 0 and 100 are defined on the *Centaur* triangular mesh from the TOSCA dataset [2]. In this figure, we display the off-diagonal entries of the inner product (grammian) matrix between the smallest 100 unit eigenvectors from the *Diag* method (Col.2, top), and the proposed discretisation (Col.2, bottom). The unit eigenvectors of the proposed discretisation are orthonormal up to machine precision, while those using the *Diag* method have an inner product of magnitudes up to 0.73. While the *lumped mass matrix* does not suffer from this drawback, we show that it lacks accuracy in the computed eigenfunctions.

The contributions of this paper are:

- We provide an FEM based discretisation of the Hamiltonian operator on triangular meshes. We prove that our discretisation yields a symmetric positive semi-definite matrix (SPSD).
- While our method guarantees an orthonormal eigenbasis for piece-wise linear functions and non-negative eigenvalues, the *Diag* method may not give orthonormal eigenbasis and may have negative, and even complex eigenvalues as demonstrated by experiments.

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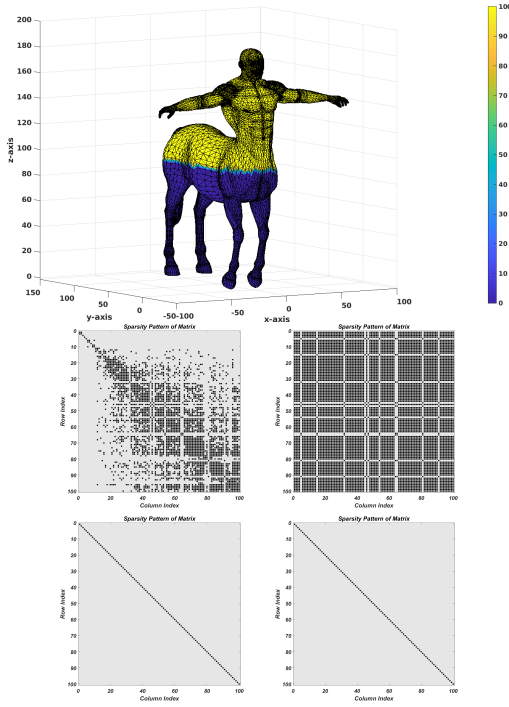


Figure 1: (Row. 1) The Centaur mesh from the TOSCA dataset with vertices colored according to the potential function. The potential is set to zero on the bottom half and to 100 on the top half. (Row. 2 & 3) Sparsity pattern of the inner product matrix for eigenvectors of the discrete Hamiltonian obtained via the *Diag* method (Row. 2) and the proposed discretisation (Row. 3). In Col. 1 (Col. 2), any entry with magnitude $\geq 10^{-3}$ (10^{-8}) is shown as a black dot. Notice that in our discretisation, the eigenvectors are orthonormal up to machine precision. Refer to Section 5 for more details.

- If the potential function is assumed to be constant, for simple geometries, Hamiltonian eigenfunctions can be computed analytically. In such cases, we show that our method gives more accurate eigenfunctions compared to the *Lumped* method. Further evidence of the accuracy of our method is given in terms of solution of the Hamiltonian version of the Poisson equation. Our method yields more accurate solutions compared to solutions obtained from the *Lumped method*. Examples are given to support our claim.

The paper is organized as follows: A brief summary of the LBO and Hamiltonian operator and their discretisation is given in Section 2, after which we derive our proposed discretisation of the Hamiltonian operator in Section 3. Properties and their proofs for the proposed discretisation are worked out in Section 4, and experiments and observations are shown in Section 5. We conclude with further directions to explore and a few questions in Section 6.

2 Related work

We briefly introduce the LBO to which the Hamiltonian operator is closely related and is the focus of this paper. It is a generalization of the Laplace operator to functions on manifolds. For any smooth surface S embedded in $\mathbb{R}^3$, the LBO

in local coordinates is given by

$$\Delta_S f = \text{div}_S(\nabla_S f) = \frac{1}{\sqrt{\det(g)}} \sum_{i,j=1}^2 \frac{\partial}{\partial x_i} \left(\sqrt{\det(g)} g^{ij} \frac{\partial}{\partial x_j} f \right), \quad (1)$$

where g_{ij} is the $(i, j)^{\text{th}}$ element of the metric (first fundamental form) on S induced from the Euclidean metric on $\mathbb{R}^3$, g^{ij} is the $(i, j)^{\text{th}}$ element of the inverse of g , and $\det(g)$ represents the determinant of the metric matrix g . The LBO can be defined on $\mathcal{L}^2$ functions using distributions: $\forall u \in \mathcal{L}^2(S), \langle 1, 2 \rangle_{\mathcal{L}^2(S)} = \langle 1, 2 \rangle_{\mathcal{L}^2(S)}, \forall \phi \in \mathcal{C}_c^\infty(S)$, where $\mathcal{C}_c^\infty(S)$ denotes the set of all smooth and real valued functions defined on a compact subset of S . We will henceforth denote the LBO by Δ , and we will use the negative of the LBO, since we wish to use its positive semi-definiteness characteristic.

A closely related operator is the Hamiltonian operator, which on a smooth surface $S \subset \mathbb{R}^3$ acts on a scalar function $f \in \mathcal{L}^2(S)$ as: $Hf = (-\Delta + V \cdot I)f$, where $V : S \rightarrow \mathbb{R}$ is a real-valued non-negative function, often known as the potential function, and I is the identity operator on $\mathcal{L}^2(S)$. Since $V \cdot I$ is a diagonal operator, the Hamiltonian is self-adjoint as well. It admits an orthonormal set of eigenfunctions which are solutions of the Helmholtz equation: $H\psi_i = E_i \psi_i, i = 1, \dots$, and forms a basis of $\mathcal{L}^2(S)$.

2.1 Discrete LBO and Hamiltonian

Following [6] we briefly describe the discretisation of the LBO operator for triangular meshes using the FEM method. Let M denote a triangular mesh that approximates the surface S . Let $\mathcal{V}$ denote the set of n vertices, $X = [x_1 \dots x_n]^T \in \mathbb{R}^{n \times 3}$ denote the vertex coordinate array, $F \in \mathbb{R}^{|F| \times 3}$ denote the face index array, $E \subset \{1, \dots, n\} \times \{1, \dots, n\}$ denote the set of edges, and T^i denote the set of triangles/faces incident on vertex i . Since function values will be specified only at the vertices, we will assume piecewise linear functions on the mesh. The vector space of all piecewise linear functions on the mesh, denoted by $P(M)$ is n -dimensional with a set of basis functions given by $B_{P(M)} = \{b_i, i = 1, \dots, n\}$ where each b_i is a piecewise linear function, also known as the top-hat function. On a triangle T_{ijk} with vertices i, j, k , the function b_i is defined as: $b_i(x) = \langle 1, 2 \rangle$, where $x^\perp$ denotes the vector obtained by rotating the vector x by 90 deg in an anticlockwise direction. The discretisation is obtained by obtaining a weak solution of the Helmholtz equation: $-\Delta f = \lambda f, f \in \mathcal{L}^2(S)$ (one can also use the Poisson equation) in $P(M)$. The weak solution $f \in P(M)$ satisfies: $\langle 1, 2 \rangle_{\mathcal{L}^2(M)} = \lambda \langle 1, 2 \rangle_{\mathcal{L}^2(M)}, i = 1, \dots, n$. Substituting $f = \sum_{i=1}^n f_i b_i(x)$ in the equation, yields the linear system of equations: $W\tilde{f} = \lambda A\tilde{f}$, where $\tilde{f} = [f_1, f_2, \dots, f_n]^T \in \mathbb{R}^n, W_{ij} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)}, 1 \leq i, j \leq n$, and $A_{ij} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)}, 1 \leq i, j \leq n$. The matrix W is often called the *stiffness matrix*, and A is called the *mass or the area matrix*. With the chosen piecewise linear basis functions, we get,

$$W_{ij} = \begin{cases} \sum_{T_k \in T^i} \frac{1}{2} (\cot \alpha_k + \cot \beta_k), & i = j, \\ -\frac{1}{2} (\cot \theta_1 + \cot \theta_2), & i \neq j \end{cases} \quad (2)$$

$$A_{ij} = \begin{cases} \sum_{T_k \in T^i} \frac{\text{Area}(T_k)}{6}, & i = j, \\ \frac{\text{Area}(T_1^{ij}) + \text{Area}(T_2^{ij})}{12}, & i \neq j \end{cases} \quad (3)$$

where α_k, β_k denote the interior angles not incident on vertex i in triangle T_k^i , and T_1^{ij}, T_2^{ij} denote the triangles sharing the edge (i, j) , and θ_1, θ_2 denote the interior angles of triangles T_1^{ij}, T_2^{ij} (resp.) opposite to the edge (i, j) . Refer to Figure 2 for a diagram. The discrete LBO spectrum consists of eigen-

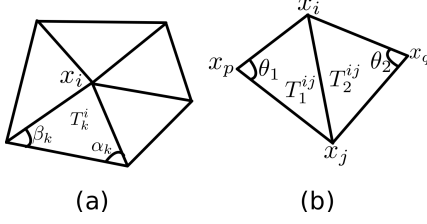


Figure 2: (a) The 1-ring neighbourhood T^i of vertex i has 5 triangles incident on vertex i . For one of these triangles T_k^i , the angles α_k, β_k are the angles opposite to vertex i in T_k^i . (b) For adjacent vertices (i, j) , there will be two triangles that share this edge, T_1^{ij}, T_2^{ij} . The angles θ_1, θ_2 are the interior angles in T_1^{ij}, T_2^{ij} (resp.) opposite edge (i, j) .

vectors and eigenvalues (e_i, λ_i) that satisfy the symmetric eigenvalue problem: $We_i = \lambda_i Ae_i, i = 1, \dots, n$. Given that A is symmetric positive definite (SPD) and W is symmetric positive semi-definite (SPSD), the eigenvectors form an A -orthonormal basis, i.e., $e_i^T Ae_j = \delta_{ij}$, and eigenvalues λ_i are non-negative. It must be noted that it was proved that any discretisation of the LBO may not satisfy all the properties satisfied by the continuous LBO [16].

The Hamiltonian operator can be discretised in a similar manner using FEM as in [4, 5, 1] by solving the Helmholtz equation $Hf = Ef$ in $P(M)$: $\langle 1, 2 \rangle_{\mathcal{L}^2(M)} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)}, i = 1, \dots, n$. With $H = -\Delta + V \cdot I$ and $f = \sum_{j=1}^n f_j b_j$, we get

$$\begin{aligned} \langle 1, 2 \rangle_{\mathcal{L}^2(M)} &= \langle 1, 2 \rangle_{\mathcal{L}^2(M)} + \langle 1, 2 \rangle_{\mathcal{L}^2(M)} \\ &= \sum_{j=1}^n \left(\langle 1, 2 \rangle_{\mathcal{L}^2(M)} + \langle 1, 2 \rangle_{\mathcal{L}^2(M)} \right) f_j \end{aligned}$$

Similarly,

$$\langle 1, 2 \rangle_{\mathcal{L}^2(M)} = E \langle 1, 2 \rangle_{\mathcal{L}^2(M)} = E \sum_{j=1}^n f_j \langle 1, 2 \rangle_{\mathcal{L}^2(M)}.$$

Using the stiffness matrix $W_{ij} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)}$, and the mass matrix $A_{ij} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)}$ (refer to Equations (2) and (3)), we get:

$$\sum_{j=1}^n \left(W_{ij} + \langle 1, 2 \rangle_{\mathcal{L}^2(M)} \right) f_j = E \sum_{j=1}^n A_{ij} f_j. \quad (4)$$

The only quantity to be worked out is $\langle 1, 2 \rangle$. The *Diag* method [1] uses the approximation $\langle 1, 2 \rangle_{\mathcal{L}^2(M)} \simeq v_j \langle 1, 2 \rangle_{\mathcal{L}^2(M)} = A_{ij} v_j$, where v_j denotes the value of the potential function at vertex j . Substituting the above approximation in Equation (4), and writing it using matrices gives:

$$(W + A \text{diag}(\bar{V})) \bar{f} = E A \bar{f}, \quad (5)$$

where $\bar{V} = [v_1, \dots, v_n]^T \in \mathbb{R}^n$ denotes the vector containing values of the potential function at the n vertices of the mesh. The discrete Hamiltonian obtained using the *Diag* method is $\hat{H}_{\text{diag}} := W + A \text{diag}(\bar{V})$.

The popular *Lumped* method on the other hand uses $\langle 1, 2 \rangle_{\mathcal{L}^2(M)} \simeq v_j \langle 1, 2 \rangle_{\mathcal{L}^2(M)} \simeq A_{lij} v_j$, where $A_{lij} = \frac{1}{3} \sum_{T_k^i \in T^i} A_{T_k^i}$ is the diagonal lumped mass matrix [13] with $A_{T_k^i}$ denoting the area of triangle T_k^i . Substituting in the Hamiltonian operator expression gives the *Lumped* Hamiltonian matrix: $\hat{H}_{\text{lump}} = W + A_l \text{diag}(\bar{V})$.

In the next section, we derive our discretisation of the Hamiltonian operator.

3 FEM-based discretisation of the Hamiltonian operator

We differ with the current discretisations in the term $\langle 1, 2 \rangle_{\mathcal{L}^2(M)}$. Since we are solving the Helmholtz equation in $P(M)$, we assume that the potential function $V \in P(M)$. Thus $V = \sum_{k=1}^n v_k b_k$, where v_k denotes the value of the potential function at vertex k . Substituting this in the above inner product gives

$$\begin{aligned} \langle 1, 2 \rangle_{\mathcal{L}^2(M)} &= \langle 1, 2 \rangle_{\mathcal{L}^2(M)} \\ &= \sum_{k=1}^n v_k \underbrace{\langle 1, 2 \rangle_{\mathcal{L}^2(M)}}_{Q_{ijk}} = \sum_{k=1}^n v_k Q_{ijk}. \end{aligned}$$

Note that $Q_{ijk} = \langle 1, 2 \rangle_{\mathcal{L}^2(M)} = \int_M b_k(x) b_j(x) b_i(x) dx = Q_{\Pi(i,j,k)}$, where $\Pi(i,j,k)$ represents any permutation of the three indices. We define a matrix $R \in \mathbb{R}^{n \times n}$ as $R_{ij} = \sum_{k=1}^n v_k Q_{ijk}$. Substituting R in Equation (4) gives us $\sum_{j=1}^n (W_{ij} + R_{ij}) f_j = E \sum_{j=1}^n A_{ij} f_j$, which when written using matrices gives us our discretisation of the Hamiltonian:

$$(W + R) \bar{f} = E A \bar{f}. \quad (6)$$

We denote this proposed discrete Hamiltonian by $\hat{H}_{\text{prop}} := W + R$. We now compute Q_{ijk} for two cases: Triangular Mesh, and an Interval in $\mathbb{R}$.

3.1 Triangular Mesh

Because of the invariance to permutation of its indices, we need to compute only Q_{iii}, Q_{iij} , for $i \neq j$, and Q_{ijk} , for $i \neq j \neq k$. Also note that since the basis functions $b_i, i = 1, \dots, n$ are non-zero only on the corresponding set of triangles $T^i, i = 1, \dots, n$, the inner product in computing Q_{ijk} can be restricted to triangles over which b_i, b_j, b_k are simultaneously non-zero. The integrand for the inner product in Q_{iii} will be non-zero only on T^i . Q_{iij} itself will be equal to zero if vertices i and j are not adjacent, since there will be no overlap between the support of b_i and b_j . If vertices i and j are adjacent, the integrand to compute Q_{iij} will be non-zero only over the two triangles incident on the edge (i, j) , i.e., over $T^i \cap T^j$. Similarly, Q_{ijk} will be zero if (i, j, k) vertices do not form a triangle of the mesh, and if they do, the integrand will be non-zero only over the triangle (i, j, k) . Thus, by direct calculations, one can derive the following non-zero terms in

$Q_{ijk}, 1 \leq i, j, k \leq n$:

$$\begin{aligned} Q_{iii} &= \int_M b_i^3(x) dx = \sum_{T_k^i \in T^i} \int_{T_k^i} b_i^3(x) dx \\ &= \frac{1}{10} \sum_{T_k^i \in T^i} \text{Area}(T_k^i), i = 1, \dots, n. \end{aligned} \quad (7)$$

$$\begin{aligned} Q_{ijj} &= \int_M b_i^2(x) b_j(x) dx \\ &= \begin{cases} \frac{1}{30} (\text{Area}(T_1^{ij}) + \text{Area}(T_2^{ij})), & \text{if } (i, j) \in E, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (8)$$

$$\begin{aligned} Q_{ijk} &= \int_M b_i(x) b_j(x) b_k(x) dx \\ &= \begin{cases} \frac{1}{60} \text{Area}(T_{ijk}), & \text{if } (i, j, k) \in F \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (9)$$

where $T_{ijk} = T^i \cap T^j \cap T^k$ denotes the triangle formed by vertices i, j and k . Note that Q_{ijk} will be a sparse 3D tensor, since for any i , only entries corresponding to some of the j, k values corresponding to vertices adjacent to vertex i will be non-zero. As a result, the matrix R with entries $R_{ij} = \sum_{k=1}^n v_k Q_{ijk}$ will also be sparse, in fact, it will be as sparse as the matrix A in the worst case. This is because only parts of the mesh having a non-zero potential value will give non-zero entries in the matrix R . Thus, as far as computational complexity is concerned, our discretisation is similar to the *Diag* method, and takes care of some of its issues.

3.2 1D Case (Interval in $\mathbb{R}$)

As a special case, we derive our discrete Hamiltonian operator on samples of an interval: $M = [a, b] \subset \mathbb{R}$. Consider the samples of the unit interval: $x_0 = a, x_1, x_2, \dots, x_n = b$, with $x_i > x_{i-1}, i = 1, \dots, n$. Note that these points are not necessarily uniformly sampled. The basis of $P([a, b])$ on these samples are $\{b_i, i = 1, \dots, n-1\}$, each defined as:

$$b_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i] \\ 1 - \frac{x - x_i}{x_{i+1} - x_i}, & x \in [x_i, x_{i+1}] \\ 0, & \text{else.} \end{cases}$$

In this case, one can readily obtain the following for $1 \leq i, j \leq n-1$:

$$Q_{ijk} = \begin{cases} \frac{x_{i+1} - x_{i-1}}{4} & i = j = k, \\ \frac{x_{i+1} - x_i}{12}, & \text{if } j = i+1, k = i \text{ or } j = i, k = i+1 \\ \frac{x_i - x_{i-1}}{12}, & \text{if } j = i-1, k = i \text{ or } j = i, k = i-1 \\ 0, & \text{else.} \end{cases} \quad (10)$$

Using Q , one can compute R , which in this case will be tridiagonal.

4 Analysis of the Discretisation

In this section, we examine the discretisations $\hat{H}_{\text{diag}}, \hat{H}_{\text{lump}}, \hat{H}_{\text{prop}}$ for properties like symmetricity, orthonormality of eigenvectors, positive semi-definiteness, and analyse the difference in the operators and their spectra.

4.1 Orthogonality of Eigenvectors

It is important to note the following induced inner product on $\mathbb{R}^n$: $\forall f, g \in P(M)$, since $f = \sum_{i=1}^n f_i b_i, g = \sum_{i=1}^n g_i b_i$, we get $\langle 1, 2 \rangle_{\mathcal{L}^2(M)} = \bar{f}^T A \bar{g}$. We refer to this inner product as the A inner product.

The continuous LBO and the Hamiltonian operator are self-adjoint on $\mathcal{L}^2(S)$ thus admitting an orthonormal basis of eigenfunctions [5]. We now prove that our discretisation $\hat{H}_{\text{prop}}$ yields an A -orthonormal set of eigenvectors. We begin by first proving that the matrix R , whose entries are defined by $R_{ij} = \sum_{k=1}^n v_k Q_{ijk}$ is SPSD.

Lemma 4.1. *The matrix $R \in \mathbb{R}^{n \times n}$ is SPSD.*

Proof. Since $R_{ij} = \sum_{k=1}^n v_k Q_{ijk} \in \mathbb{R}$, and Q is invariant to permutation of its indices, it follows that R is symmetric:

$$R_{ji} = \sum_{k=1}^n v_k Q_{jik} = \sum_{k=1}^n v_k Q_{ijk} = R_{ij}.$$

Let $p \in \mathbb{R}^n$. Then

$$\begin{aligned} p^T R p &= \sum_{i,j=1}^n p_i R_{ij} p_j = \sum_{i,j=1}^n p_i p_j \sum_{k=1}^n v_k \int_{\Omega} b_i(x) b_j(x) b_k(x) dx \\ &= \sum_{k=1}^n v_k \int_{\Omega} \left(\sum_{i=1}^n p_i b_i(x) \right) \left(\sum_{j=1}^n p_j b_j(x) \right) b_k(x) dx \end{aligned}$$

Letting $p(x) = \sum_{i=1}^n p_i b_i(x)$ gives,

$$p^T R p = \sum_{k=1}^n v_k \int_{\Omega} (p(x))^2 b_k(x) dx.$$

Since $v_k \geq 0$, and $\forall x \in \Omega, p(x)^2 \geq 0, b_k(x) \geq 0$, we have that $p^T R p \geq 0$, which proves the positive semi-definiteness of R . $\square$

Since the matrix W is real symmetric, our discrete Hamiltonian matrix $\hat{H}_{\text{prop}} = W + R$ is real symmetric as well. Using this result, we now prove that $\hat{H}_{\text{prop}}$ yields an orthonormal set of eigenvectors.

Theorem 4.2. *Let $\hat{H}_{\text{prop}} \bar{f} = \lambda A \bar{f}$ be the generalized symmetric eigenvalue problem. Given that $\hat{H}_{\text{prop}}$ is real symmetric, and the mass matrix A is SPD, the n (generalized) eigenvalues $\lambda_1, \dots, \lambda_n$ are real. Moreover the corresponding eigenvectors $\{\bar{e}_1, \dots, \bar{e}_n\}$ form an A -orthonormal basis of $\mathbb{R}^n$, i.e., $\bar{e}_i^T A \bar{e}_j = \delta_{ij}, 1 \leq i, j \leq n$.*

Proof. First we will show that the roots of the characteristic equation $\det(\hat{H}_{\text{prop}} - \lambda A) = 0$ are all real. Assume that λ_i is a root of this equation. Then using the Cholesky decomposition of the SPD matrix A : $A = LL^T$, we have $\det(\hat{H}_{\text{prop}} - \lambda_i LL^T) = 0$ iff $\det(L^{-1} \hat{H}_{\text{prop}} (L^T)^{-1} - \lambda_i I) = 0$. Therefore, λ_i is a root of the characteristic equation $\det(\hat{H}_{\text{prop}} - \lambda A) = 0$ if and only if λ_i is an eigenvalue of the matrix $\hat{H}_{\text{prop}}^L = L^{-1} \hat{H}_{\text{prop}} (L^T)^{-1}$. By a quick calculation, we can see that $\hat{H}_{\text{prop}}^L$ is real symmetric, and therefore all its eigenvalues are real. This completes the proof of the first part of the theorem.

Note that since $\hat{H}_{\text{prop}}^L$ is real symmetric, it has an orthonormal set of eigenvectors, say $\{\bar{u}_i, i = 1, \dots, n\}$, where orthogonality is defined using the Euclidean inner product. Define

$\bar{e}_i = (L^T)^{-1} \bar{u}_i, i = 1, \dots, n$. Observe that for $i = 1, \dots, n$,

$$\begin{aligned}\hat{H}_{\text{prop}}^L \bar{u}_i &= \lambda_i \bar{u}_i \\ L^{-1} \hat{H}_{\text{prop}} (L^T)^{-1} \bar{u}_i &= \lambda_i \bar{u}_i \\ (L^T)^{-1} L^{-1} \hat{H}_{\text{prop}} (L^T)^{-1} u_i &= \lambda_i (L^T)^{-1} u_i \\ (LL^T)^{-1} \hat{H}_{\text{prop}} \bar{e}_i &= \lambda_i \bar{e}_i \\ \hat{H}_{\text{prop}} \bar{e}_i &= \lambda_i A \bar{e}_i.\end{aligned}$$

Thus, $\bar{e}_i = (L^T)^{-1} \bar{u}_i, i = 1, \dots, n$ are generalized eigenvectors of $(\hat{H}_{\text{prop}}, A)$. Finally, we show A -orthonormality of $\bar{e}_i, i = 1, \dots, n$:

$$\begin{aligned}\bar{e}_i^T A \bar{e}_j &= \left((L^T)^{-1} \bar{u}_i \right)^T (LL^T) (L^T)^{-1} \bar{u}_j \\ &= \bar{u}_i^T L^{-1} LL^T (L^T)^{-1} \bar{u}_j = \bar{u}_i^T \bar{u}_j = \delta_{ij},\end{aligned}$$

where the last equality is due to orthonormality of $\{\bar{u}_i, i = 1, \dots, n\}$. $\square$

In the discretisation $\hat{H}_{\text{diag}} = W + A \text{diag}(\bar{V})$, note that the matrix $A \text{diag}(\bar{V})$ is not necessarily symmetric, thus the eigenvectors of $\hat{H}_{\text{diag}}$ may not be orthonormal. Moreover $\hat{H}_{\text{diag}}$ is not positive semi-definite either. We show several examples where the eigenvectors of $\hat{H}_{\text{diag}}$ are not orthonormal, and an example where $x^T \hat{H}_{\text{diag}} x < 0$ in Section 5. Note that the matrix $\hat{H}_{\text{lump}} = W + A_l \text{diag}(\bar{V})$ is SPSPD, since A_l and $\text{diag}(\bar{V})$ are diagonal matrices with positive and non-negative entries resp. This ensures that $\hat{H}_{\text{lump}}$ yields A_l -orthonormal eigenvectors.

We now show that the discretisations $\hat{H}_{\text{lump}}, \hat{H}_{\text{prop}}$ yield non-negative eigenvalues, thus giving ideal spectral properties.

Lemma 4.3. *If the discrete LBO cotan matrix W is SPSPD, the generalized eigenvalues of $\hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$ are non-negative.*

Proof. Since W and R are SPSPD, $\hat{H}_{\text{prop}} = W + R$ is also SPSPD. Similarly, since $A_l, \text{diag}(\bar{V})$ are diagonal with positive and non-negative entries, $\hat{H}_{\text{lump}} = W + A_l \text{diag}(\bar{V})$ is also SPSPD. For any generalized eigenvalue-(unit) eigenvector pair $(\lambda, \bar{e})$ of $\hat{H}_{\text{prop}}$, we have $\hat{H}_{\text{prop}} \bar{e} = \lambda A \bar{e}$. Multiplying both sides by $\bar{e}^T$ gives $\bar{e}^T \hat{H}_{\text{prop}} \bar{e} = \lambda \bar{e}^T A \bar{e} = \lambda$. Since $\hat{H}_{\text{prop}}$ is SPSPD, $\lambda = \bar{e}^T \hat{H}_{\text{prop}} \bar{e} \geq 0$, proving that all eigenvalues of $\hat{H}_{\text{prop}}$ are non-negative. Replacing $\hat{H}_{\text{prop}}, A$ by $\hat{H}_{\text{lump}}, A_l$ above proves that all eigenvalues of $\hat{H}_{\text{lump}}$ are non-negative. $\square$

Next we give a bound on the differences in eigenvectors of $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ considering $\hat{H}_{\text{diag}}$ as a perturbation of $\hat{H}_{\text{prop}}$.

4.2 Perturbation Analysis

We can write $\hat{H}_{\text{diag}} = \hat{H}_{\text{prop}} + \delta R$, where $\delta R = A \text{diag}(\bar{V}) - R$. Let $\lambda_i, i = 1, \dots, n$ and $\bar{e}_i, i = 1, \dots, n$ denote the eigenvalues and eigenvectors of $\hat{H}_{\text{prop}}$. We will examine the perturbation of the eigenvalues and eigenvectors of $\hat{H}_{\text{prop}}$ when perturbed by δR to obtain eigenvalues and eigenvectors of $\hat{H}_{\text{diag}}$: $(\lambda_i + \delta \lambda_i, \bar{e}_i + \delta \bar{e}_i)$. With an assumption that the eigenvalues of $\hat{H}_{\text{prop}}$ are simple, one can derive that the perturbation in the eigenvectors and eigenvalues (up to first order) are:

$$\delta \bar{e}_i = \sum_{j \neq i, 1}^n \frac{\bar{e}_j^T \delta R \bar{e}_i}{(\lambda_i - \lambda_j)} \bar{e}_j, \text{ and } \delta \lambda_i = \bar{e}_i^T \delta R \bar{e}_i \quad (11)$$

The perturbation in the spectra depends on the matrix $\delta R = A \text{diag}(\bar{V}) - R$ which is analysed next.

4.3 Necessary and Sufficient Condition for

$$\hat{H}_{\text{diag}} = \hat{H}_{\text{prop}}$$

We present necessary and sufficient conditions for $\hat{H}_{\text{diag}} = \hat{H}_{\text{prop}}$ below.

Theorem 4.4. *The two-discretisations $\hat{H}_{\text{diag}} = \hat{H}_{\text{prop}}$ of the Hamiltonian agree if and only if the potential function V is constant, assuming zero boundary conditions on the potential if the mesh has a boundary.*

Proof. For $\hat{H}_{\text{diag}} = \hat{H}_{\text{prop}}$, we must have $R = A \text{diag}(\bar{V})$. Then, using definitions of R, A, Q_{iii}, Q_{ijj} , we derive conditions for $R - A \text{diag}(\bar{V}) = 0$ below:

$$(R - A \text{diag}(\bar{V}))_{ii} = \sum_{k=1}^n v_k Q_{iik} - A_{ii} v_i = 0$$

$$v_i Q_{iii} + \sum_{j \in N(i)} v_j Q_{ijj} - v_i \left(\frac{1}{6} \sum_k \text{Area}(T_k^i) \right) = 0$$

$$\begin{aligned}v_i \left(\frac{1}{10} \sum_k \text{Area}(T_k^i) \right) + \sum_{j \in N(i)} v_j \left(\frac{\text{Area}(T_1^{ij}) + \text{Area}(T_2^{ij})}{30} \right) \\ - v_i \left(\frac{1}{6} \sum_k \text{Area}(T_k^i) \right) = 0,\end{aligned}$$

where T_1^{ij}, T_2^{ij} are the two triangles incident on the edge (i, j) . Simplifying the above expression, gives:

$$\sum_{j \in N(i)} (v_j - v_i) \left(\frac{\text{Area}(T_1^{ij}) + \text{Area}(T_2^{ij})}{30} \right) = 0.$$

$$\frac{2}{5} \sum_{j=1}^n A_{ij} (v_j - v_i) = 0 \quad (12)$$

$$v_i = \frac{\sum_{j=1, j \neq i}^n A_{ij} v_j}{\sum_{j=1, j \neq i}^n A_{ij}} \quad (13)$$

where A denotes the area/mass matrix defined in Equation (3). Since $\sum_{j=1, j \neq i}^n A_{ij} = A_{ii}$, a constant potential $v_k = v \in \mathbb{R}, k = 1, \dots, n$ satisfies Equation (12). Since $A_{ij} \geq 0, v_i \geq 0, i = 1, \dots, n$, Equation (13) implies that the potential values at each vertex must be equal to a convex combination of potential values at the neighboring vertices (harmonic function). The potential function values can have a maximum only at the boundary. For a mesh without boundary, the potential must thus be constant, while for a mesh with boundary and assumed zero boundary conditions (for the potential), the potential must be equal to zero everywhere. $\square$

Since a constant potential function in the Hamiltonian operator will yield the same eigenvectors as those of the LBO and eigenvalues shifted by the same constant, such a potential function is of no practical use. Thus, the two discretisations $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$, will always differ in practice and yield different spectra. While larger variations in the potential will lead to a larger δR (larger difference between $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$), this may or may not impact the non-orthogonality of eigenvectors of $\hat{H}_{\text{diag}}$.

4.4 Computational Complexity and Sparsity

We compare the computational complexity and required memory for approximations $\hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$. Let d_m denote

the maximum degree of any vertex in a mesh with n vertices. In order to compute the matrix R , assuming that the triangle areas have already been computed (in order to compute the Cotan Laplacian for instance) the number of floating point operations per vertex is bounded above by $14 + 5d_m$. Thus the computational complexity for the matrix R for a mesh is $\mathcal{O}(nd_m)$. In order to compute the lumped mass matrix A_l , the number of operations per vertex is at most d_m , giving a total computational cost of nd_m . Computing the matrix-vector product, $A_l \text{diag}(\tilde{V})$ requires an additional n operations. Hence, the overall complexity is $\mathcal{O}(n(d_m + 1)) \approx \mathcal{O}(nd_m)$. Some timing results in MATLAB are given in Section 5.

As far as memory consumption is concerned, the matrix R has at most $n(d_m + 1)$ nonzero elements, and with all non-zero entries overlapping with the non-zero entries of the Cotan Laplacian. Hence, the matrix $\hat{H}_{\text{prop}}$ stores at most $n(d_m + 1)$ nonzero elements. This is exactly the case also with the Hamiltonian approximation $\hat{H}_{\text{lump}}$.

5 Experiments

In this section, we provide experimental results that capture the difference in the three discretisations: $\hat{H}_{\text{diag}}$, $\hat{H}_{\text{lump}}$, $\hat{H}_{\text{prop}}$. In Subsection 5.1, we show non-orthogonality in eigenvectors of $\hat{H}_{\text{diag}}$ on the interval, as well as on meshes with different choices for the potential function. In Subsection 5.2, we show errors in the three schemes in approximating eigenvalues and eigenfunctions of the Hamiltonian operator, while we analyse errors in solution of the Hamiltonian-based Poisson equation in the three schemes in Subsection 5.3. All experiments have been performed on an Intel i7 (13th gen) laptop with 16GB RAM using MATLAB 2024a.

5.1 Orthogonality of Eigenfunctions

We compare the spectra of $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ on (1) an interval of $\mathbb{R}$, under both uniform and non-uniform sampling, and (2) triangular meshes. The comparison is given in terms of the inner products between the corresponding sets of eigenvectors. Let n denote the number of vertices (excl. vertices at boundary) in the interval or a mesh, as the case may be, and let $\bar{\Psi}_{\text{diag}}, \bar{\Psi}_{\text{prop}} \in \mathbb{R}^{n \times k}$ denote the matrices whose $k \leq n$ columns consist of unit-eigenvectors of $\hat{H}_{\text{diag}}, \hat{H}_{\text{prop}}$ resp. We compare the number of non-zero terms in the inner-product/Grammian matrices: $P_{\text{diag}} := \bar{\Psi}_{\text{diag}}^T A \bar{\Psi}_{\text{diag}}, P_{\text{prop}} := \bar{\Psi}_{\text{prop}}^T A \bar{\Psi}_{\text{prop}}$, where A denotes the mass/area matrix. Ideally, we expect the inner product matrix to be exactly equal to the $k \times k$ identity matrix. Numerically, we examine the matrices by counting the number of entries with magnitude greater than or equal to the thresholds: $[1e - 10, 1e - 9, \dots, 1e - 1]$. Ideally, this should always be equal to k . We plot the number of entries with magnitude greater than these thresholds against the logarithm (base 10) of the corresponding thresholds for both P_{diag} and P_{prop} to visualize the difference. We will refer to such a plot as *Sparsity pattern*. The generalized eigenvector-eigenvalue pairs of the form $He = \lambda Ae$ are computed using `eig(H,A)` for small matrices, and using `eigs(H,A,k,'smallestabs')` for large and sparse matrices coming from meshes, with $k = 100$.

1. *Interval $[a, b] \subset \mathbb{R}$* : We show examples of the discretised Hamiltonian on both, uniformly and non-uniformly sampled 10 points (incl. 0 and 10) in the interval $[0, 10]$, for various potential functions: step, linear (or smooth) and random. Note that we use zero-boundary conditions, i.e., all functions f considered on $[0, 10]$ satisfy: $f(0) = f(10) = 0$. We get a set of 8 eigenvectors for both discretisations in each case, and the matrices $P_{\text{diag}}, P_{\text{prop}} \in \mathbb{R}^{8 \times 8}$. For the uniformly sampled case, these potential functions and the corresponding sparsity pattern plots for the two inner product matrices $P_{\text{diag}}, P_{\text{prop}}$ are shown in Figure 3. Observe that for all 3 potential functions considered, the sparsity pattern in P_{prop} is ideal, i.e., exactly 8 (diagonal) of the possible 64 entries have magnitude greater than $1e - 10$. Similar simulations

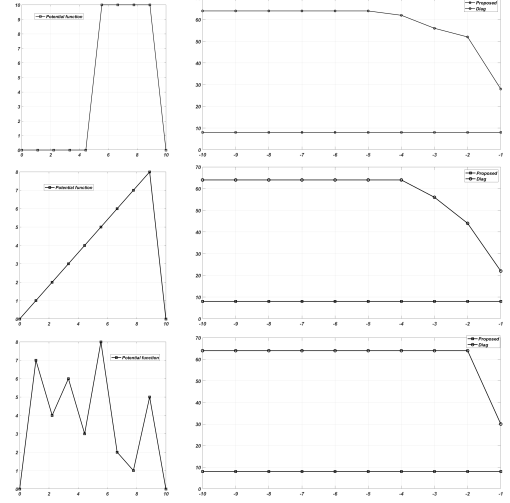


Figure 3: Experiments on uniformly sampled 10 points in $[0, 10]$. (Row 1) Step potential function and the sparsity pattern, (Row 2) Linear potential and the sparsity pattern, (Row 3) Random (High variation) potential and sparsity pattern. The values of the potential function at the sampled points are marked by a $\square$, and the sparsity pattern on P_{diag} is shown via the $\circ$ marker, while that of P_{prop} is shown via the $\square$ marker. In all cases, our discretisation shows an ideal behavior.

and plots for non-uniformly sampled points on the interval $[0, 10]$ are given in Figure 4. For our discretisation $\hat{H}_{\text{prop}}$, the eigenvectors remain orthonormal to a large extent, i.e., the inner product matrix P_{prop} has exactly 8 entries with magnitude greater than equal to $1e - 10$, while all entries in P_{diag} have a magnitude greater than $1e - 5$ for all the six experiments counting the uniformly sampled case. Notice the behavior of $\hat{H}_{\text{diag}}$ deteriorates in the case of random potential function, and is the best when the potential varies slowly (linear potential case).

2. *Triangular Meshes*: For a step potential (values 0 and 100) defined on the *Centaur* mesh (# of vertices = 15768, # of faces = 31532) from the TOSCA dataset[2], we compute the first (smallest eigenvalue) 100 eigenvectors. The sparsity pattern of the inner product matrices $P_{\text{diag}}, P_{\text{prop}} \in \mathbb{R}^{100 \times 100}$ are shown in Figure 1 for two thresholds $1e - 3$ (Col. 1) and $1e - 8$ (Col. 2). While for the proposed discretisation $\hat{H}_{\text{prop}}$ (bottom row) only the diagonal entries of P_{prop} have magnitudes above these thresholds, 730 entries in P_{diag} (top row) have magnitudes greater than or equal to $1e - 3$ and all but two magnitudes (9998) greater than or equal to $1e - 8$. This

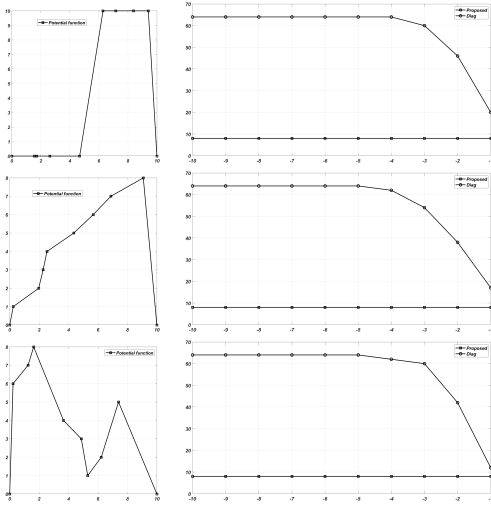


Figure 4: Experiments on non-uniformly sampled 10 points in $[0, 10]$. (Row 1) Step potential function and the sparsity pattern, (Row 2) Linear potential and the sparsity pattern, (Row 3) Random (High variation) potential and sparsity pattern. The values of the potential function at the sampled points are marked by a $\square$, and the sparsity pattern on P_{diag} is shown via the $\circ$ marker, while that of P_{prop} is shown via the $\square$ marker. In all cases, our discretisation shows an ideal behavior.

demonstrates the non-orthogonality of eigenvectors of $\hat{H}_{diag}$ and orthogonality of eigenvectors of $\hat{H}_{prop}$. In fact, all off-diagonal entries of P_{prop} are in the range of $1e - 14$. We obtain similar results on other meshes with different potential functions. We repeat this experiment on the Stanford Bunny mesh ($\sim 55K$ vertices, $\sim 110K$ faces), and the Armadillo mesh [11] ($\sim 170K$ vertices, $\sim 350K$ faces). We use a smooth potential function between 0 and 1 on the Stanford Bunny mesh, and a square wave like potential on the Armadillo mesh. For both meshes, the inner product matrix P_{diag} , obtained from eigenvectors of $\hat{H}_{diag}$, has several values with magnitude greater than $1e - 4$, while all its entries have a magnitude greater than $1e - 7$ in the Stanford Bunny case, and about 7500 of them greater than $1e - 7$ in the Armadillo mesh. For the inner product matrix P_{prop} obtained from eigenvectors of $\hat{H}_{prop}$, there are no off-diagonal entries with magnitude greater than even $1e - 10$. We then assign samples generated from a uniform random variable $U[0, 10]$ to the top half of the Armadillo mesh, keeping the potential for the bottom half at zero. The sparsity pattern is shown in Figure 6. We perform a similar experiment for all 17¹ meshes in the SHREC’19 dataset [7]. With a step potential (bottom half zero, top half 100), the sparsity pattern are shown in Figure 7.

3. *Positive Semidefiniteness*: As proved in Section 4, the matrices $\hat{H}_{lump}$, $\hat{H}_{prop}$ are symmetric and positive semi-definite as far as the Cotan matrix W is positive semi-definite. The same cannot be said about $\hat{H}_{diag}$. Since $\hat{H}_{diag}$ is not symmetric, we discuss positive definiteness of $\hat{H}_{diag}$ using eigenvalues of $\hat{H}_{diag}^{sym} := \frac{1}{2}(\hat{H}_{diag} + \hat{H}_{diag}^T)$. Because $x^T \hat{H}_{diag} x = x^T \hat{H}_{diag}^{sym} x$, if there are negative eigenvalues of $\hat{H}_{diag}^{sym}$, the matrix $\hat{H}_{diag}$ is not positive semi-definite. For the potential function shown in

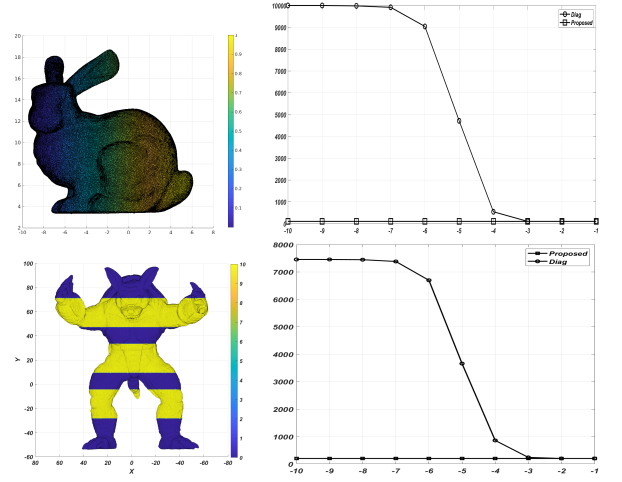


Figure 5: (Left column) The Stanford Bunny mesh (top), and the Armadillo mesh (bottom), colored according to the potential function. (Right column) The sparsity pattern. The marker $\circ$ is used in the sparsity pattern of $\hat{H}_{diag}$ (matrix P_{diag}), while the marker $\square$ is used for the sparsity pattern of $\hat{H}_{prop}$ (matrix P_{prop}). Notice that for both meshes, the number of entries with magnitude larger than any of the given threshold remains fixed at the ideal number: 100 for our discretisation.

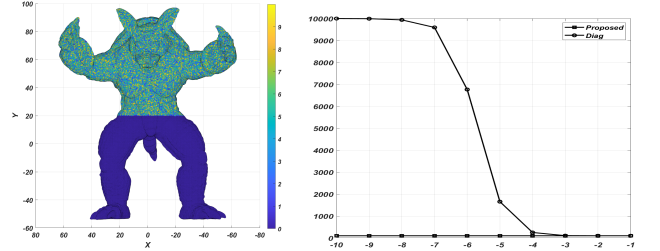


Figure 6: (left) The Armadillo mesh with colour according to the potential function. (right) The sparsity pattern for the inner product matrices P_{diag} (marker - $\circ$) and P_{prop} (marker - $\square$). Observe the larger non-negligible entries in P_{diag} . Compare with Figure 5 (bottom-right)

Figure 8(left), the smallest generalized eigenvalue of $\hat{H}_{diag}^{sym}$ is -1.5623 , while that of our discretisation $\hat{H}_{prop}$ is 0.9026 . The generalized eigenvector of $\hat{H}_{diag}$ corresponding to the negative eigenvalue is shown in Figure 8(right).

5.2 Error in Eigenvalues and Eigenvectors

We now examine the approximation error between eigenvalues and eigenvectors (eigenfunctions) of the continuous Hamiltonian operator H and those computed via the three discretisations. For an arbitrary potential function V on any arbitrary domain S (interval or surface mesh), it may not be possible to solve the equation: $H\psi = \lambda\psi$ analytically. For the purpose of comparison, we choose constant potential function $V \equiv k > 0$ on the (1) unit interval $[0, 1] \subset \mathbb{R}$, and (2) unit sphere triangular mesh, for which analytical expressions of the eigenfunctions are well known. In this case, the eigenfunctions of the Hamiltonian will be those of the LBO, while the eigenvalues will differ from those of the LBO exactly by k , the constant value of the potential function.

¹We used all manifold meshes and meshes without boundaries

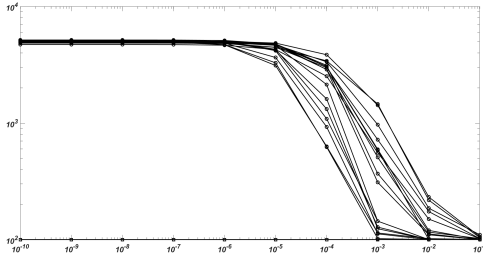


Figure 7: Experiments over 17 meshes on SHREC'19 [7] data set. The error threshold is given on the x-axis, while the number of non-zero entries in P_i , $i = 1, 3$ are shown on the y-axis. The sparsity pattern for the inner product matrices P_1 (marker $\circ$) and P_3 (marker $\square$).

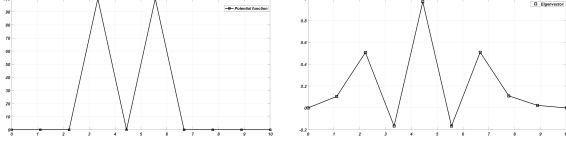


Figure 8: (left) An example of a potential function on uniformly sampled 10 points on $[0, 10]$ which leads to an indefinite matrix $\hat{H}_{\text{diag}}$. (right) The unit eigenvector of $\hat{H}_1^{\text{sym}}$ corresponding to the negative eigenvalue (-1.5623) .

In the unit interval case, we assume zero-boundary conditions. For the unit sphere case, we use spherical coordinates $(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$ to express the Hamiltonian operator, and solve for the eigenfunctions (known as *spherical harmonics*) and eigenvalues. The eigenvalues and eigenfunctions of the Hamiltonian with constant potential $k > 0$ for the unit interval are:

$$\lambda_l = l^2\pi^2 + k, \quad \psi_l(x) = \sin(l\pi x), l \geq 1, x \in [0, 1], \quad (14)$$

while the eigenvalues and eigenfunctions (real spherical harmonics) for the unit sphere are:

$$\lambda_l = l(l+1) + k, l \geq 0 \quad (15)$$

$$\psi_l^m(\theta, \phi) = \begin{cases} \sqrt{2}K_l^m \cos(m\phi) P_l^m(\cos\theta) & \text{if } m > 0, \\ K_l^0 P_l^0(\cos\theta) & \text{if } m = 0, \\ \sqrt{2}K_l^m \sin(-m\phi) P_l^{-m}(\cos\theta) & \text{if } m < 0. \end{cases} \quad (16)$$

where $l \geq 0, -l \leq m \leq l$, K_l^m are the normalization constants: $K_l^m = \sqrt{\frac{(2l+1)!}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}}$, and P_l^m are the associated Legendre polynomials. Note that for the unit sphere, each eigenvalue $\lambda_l, l \geq 0$ occurs with a multiplicity of $2l+1$. The analytical eigenvalues and those obtained via matrices $\hat{H}_{\text{diag}}, \hat{H}_{\text{lump}}, \hat{H}_{\text{prop}}$ for the unit interval case are shown in Figure 9, and for the unit sphere case are shown in Figure 10. The matrices $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ are exactly the same except for values from points near boundaries. This difference arises due to potential being zero at the boundary, and the difference has almost no effect on the eigenvalues. As can be observed in the eigenvalue plots for the unit interval and the mesh in Figures 9 and 10, the eigenvalues for $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ coincide. The eigenvalues of $\hat{H}_{\text{lump}}$ underestimate those of the continuous Hamiltonian, while eigenvalues of $\hat{H}_{\text{diag}}, \hat{H}_{\text{prop}}$ overestimate them. In the case of unit sphere (Figure 10), note that the spectral gap is better preserved by $\hat{H}_{\text{diag}}, \hat{H}_{\text{prop}}$

compared to $\hat{H}_{\text{lump}}$. This may lead to less stable behavior of the eigenspaces in $\hat{H}_{\text{lump}}$ compared to the eigenspaces of $\hat{H}_{\text{diag}}, \hat{H}_{\text{prop}}$ [8]. Verifying this aspect in a general case (arbitrary surface and potential function) may not be possible though.

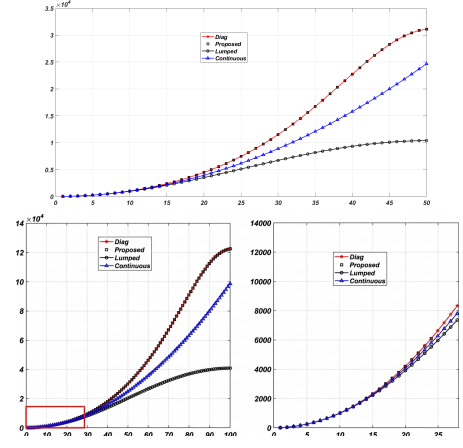


Figure 9: Eigenvalues of continuous Hamiltonian and those of $\hat{H}_{\text{diag}}, \hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$. Eigenvalues of $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ coincide and overestimate the true eigenvalues, while eigenvalues of $\hat{H}_{\text{lump}}$ underestimate the true eigenvalues. (Row 1) Eigenvalues for 50 point discretisation of unit interval. (Row 2) 100 point discretisation of unit interval, and the corresponding smallest 28 eigenvalues.

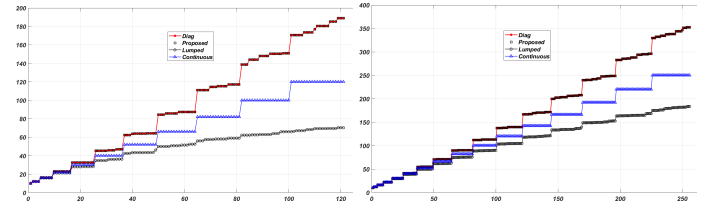


Figure 10: Eigenvalues of continuous Hamiltonian with constant potential and the three discretisations $\hat{H}_{\text{diag}}, \hat{H}_{\text{lump}}, \hat{H}_{\text{prop}}$ on the unit sphere. The eigenvalues for $\hat{H}_{\text{diag}}$ and $\hat{H}_{\text{prop}}$ coincide, and overestimate the true eigenvalues, while eigenvalues of $\hat{H}_{\text{lump}}$ underestimates the true values. The smallest 121(left) and 256(right) eigenvalues of a sphere with 162 and 642 vertices resp. are shown.

We next compare the errors in the eigenvectors for $\hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$ for the unit interval and unit sphere case. The plot for the L^2 error for the unit interval case for the first, third, fifth and seventh (in ascending order of eigenvalues) eigenfunction against the number of points sampled on the interval is shown in Figure 11. While both methods yield eigenvectors that converge to the true eigenfunctions, the error in the proposed method is consistently lower. Moreover, the gap in the errors increases as we observe eigenvectors corresponding to larger eigenvalues.

In case of unit sphere, eigenspaces of dimension greater than 1 are associated with all but one eigenvalue ($l = 0$). In order to compare eigenfunctions, for each eigenfunction of the continuous Hamiltonian, we least-squares fit a linear combination of eigenvectors from a corresponding eigenspace of the discretised Hamiltonians $\hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$, before computing the L^2 error on the vertices of the sphere.

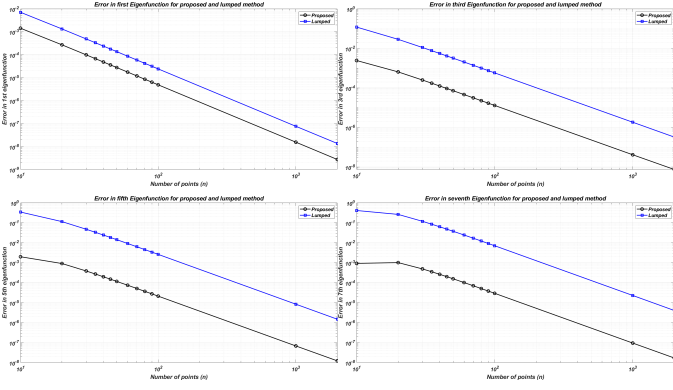


Figure 11: Error in estimate of eigenfunctions of continuous Hamiltonian operator on $[0, 1] \subset \mathbb{R}$. The $\log$ of the L^2 error is shown against the number of equidistant sampled points ($\{10, 20, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 1000, 2000\}$) on $[0, 1]$, for the 1st, 3rd, 5th and 7th eigenfunction in ascending order of eigenvalues. Increasing the number of points decreases the error, while the error gap between eigenfunctions estimated through the proposed and lumped method increases.

The plot for the L^2 error between the first few eigenfunctions ($l = 0, \dots, 5, m \leq l$) of the continuous Hamiltonian and least-squares fit eigenvectors from the corresponding eigenspace of $\hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$ against mesh resolution is shown in Figure 12. While for $l = 0$ (eigenfunction corresponding to smallest eigenvalue) the errors in the computed eigenvectors through $\hat{H}_{\text{lump}}, \hat{H}_{\text{prop}}$ are of the order 10^{-10} or lower, with those computed using $\hat{H}_{\text{lump}}$ slightly better; the difference in the errors is close to numerical precision (10^{-13}). For higher eigenfunctions, the errors in eigenvectors increase in both cases, and errors from eigenvectors computed from $\hat{H}_{\text{prop}}$ is lower than the errors for eigenvectors from $\hat{H}_{\text{lump}}$.

5.3 Poisson equation with Hamiltonian operator

We now compare the analytical solution of the Poisson equation with Hamiltonian operator and homogeneous Dirichlet boundary conditions on an interval $[a, b] \subset \mathbb{R}$, with discrete approximations obtained from $\hat{H}_{\text{diag}}, \hat{H}_{\text{lump}}$ and $\hat{H}_{\text{prop}}$. The Poisson equation using the Hamiltonian operator with zero boundary conditions on an interval $[a, b] \subset \mathbb{R}$ is defined as:

$$-u''(x) + V(x)u(x) = f(x), \quad x \in [a, b], \quad \text{with } u(a) = u(b) = 0.$$

In what follows, we refer to this equation as the *Poisson equation*. The corresponding weak formulation: $\int_a^b [u'(x)\varphi'(x) + V(x)u(x)\varphi(x)]dx = \int_a^b f(x)\varphi(x)dx, \forall \varphi \in H_0^1(a, b)$ has a unique solution for any non-negative and bounded potential function V , and $f \in L^2(a, b)$. For a proof, refer to A.

As before, we discretise this equation using piece-wise linear functions. Restricting the solution, potential and test functions to the vector space of piece-wise linear functions: $u(x) = \sum_{k=1}^n u_k b_k(x), V(x) = \sum_{k=1}^n v_k b_k(x)$, gives the left hand side as $\hat{H}_{\text{diag}}\bar{u}, \hat{H}_{\text{lump}}\bar{u}$ or $\hat{H}_{\text{prop}}\bar{u}$ depending on the approximation used, where $\bar{u} \in \mathbb{R}^n$ denotes the vector of coefficients of the solution u in the piece-wise linear basis. The right hand side is the vector: $\bar{f} =$

$\left(\int_a^b f(x)b_1(x)dx, \dots, \int_a^b f(x)b_n(x)dx\right)^T \in \mathbb{R}^n$. Since this is the orthogonal projection of f to the space of piece-wise linear functions on $[a, b]$, there exists $c \in \mathbb{R}^n$ such that $Ac = \bar{f}$. When using the lumped mass matrix A_l to approximate the mass or the Gram matrix of the basis $\{b_k, k = 1, \dots, n\}$, we should use the same approximation on the right hand side: $\left(\int_a^b f(x)b_1(x)dx, \dots, \int_a^b f(x)b_n(x)dx\right)^T = Ac \simeq A_l c$. Thus the discretised Poisson equations when using the *Diag*, *Lumped*, and proposed scheme resp. are:

$$\hat{H}_{\text{diag}}\bar{u} = Ac, \quad \hat{H}_{\text{lump}}\bar{u} = A_l c, \quad \hat{H}_{\text{prop}}\bar{u} = Ac \quad (17)$$

In examples that we have worked with, we use polynomials, trigonometric functions, and exponentials for f , for which the vector $\bar{f}$ can be evaluated analytically. From $\bar{f}$, we then calculate $c = A^{-1}\bar{f}$ that is used on the right hand sides in the equations above.

We give an example where the potential function is assumed to be constant $V \equiv k > 0$, in which case the analytical solutions to the Poisson equation can be found. In general, solutions exist for a much wider class of potential functions, but computing them may not be possible. We show one representative example with $\Omega = [0, 1] \subset \mathbb{R}$, f a combination of polynomial, trigonometric and exponential functions: $f = \sin(x) + e^x + x^3$, and potential $V \equiv 4$. The analytical solution is given as:

$$u(x) = -\frac{1}{3} - \left(\frac{-e^2}{3} + \frac{e}{3} + \frac{\sin(1)}{5} + \frac{5}{8}\right) e^{2x} + \frac{e^x}{3} + \frac{\sin(x)}{5} + \left(\frac{-e^2}{3} + \frac{e}{3} + \frac{\sin(1)}{5} + \frac{5}{8}\right) e^{-2x} + \frac{2x^3 + 3x}{8}$$

The analytical solution, and the three discrete approximations obtained by solving Equation 17 are shown in Figure 13 on 4 and 8 points in the interval $[0, 1]$. The solutions appear to align, showing signs of convergence. In what follows, by errors in the solution, we mean the average per point error: $\frac{1}{n} \|u_{\text{analytical}} - u_{\text{numerical}}\|_2$, where n is the number of samples in case of interval, and number of vertices in case of mesh. The errors in the discretisations against the number of sampled points for this example as well as for $f(x) = \sin(x) + e^x + x^3$ and $V(x) = \exp(x)$ is shown in Figure 14. It can be observed that average per point error is lower in the proposed and *Diag* method compared to the *Lumped* method, with a faster reduction in error as the number of points is increased. Similar behavior is observed for the case when the right hand side function f is a polynomial, trigonometric and exponential function separately. A couple of examples with a non-constant but continuous potential, on the interval, and on the unit sphere (in this case, the right hand side is computed numerically) are shown in Appendix B.

5.4 Computational Complexity:

In Table 1, the average computation time (in secs) and its standard deviation for 100 experiments for assembling the matrices and computing the eigenvalues and eigenvectors are given for four meshes with different numbers of vertices. Table 2 reports the same for the Poisson equation over an interval and sphere meshes with different resolution. The recorded times include assembling the matrices and computing the numerical solution. The functions

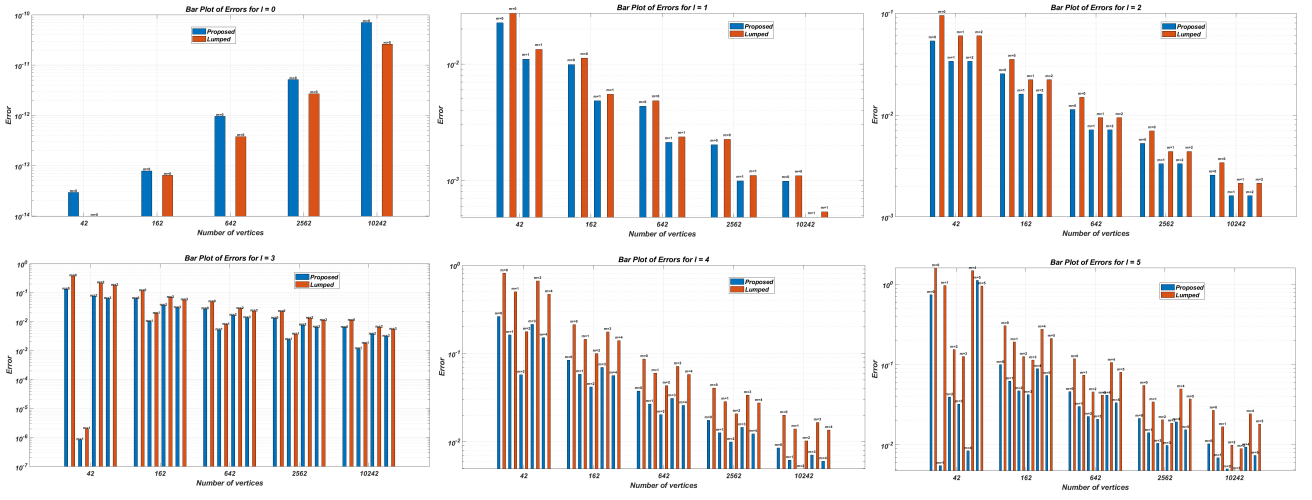


Figure 12: Experiments on the error in eigenfunctions with the proposed and lumped method with respect to the spherical harmonic functions. The number of vertices on the sphere mesh varies along the x-axis, and the error is shown on the y-axis. The errors in the proposed method are shown in blue, and the error in the *Lumped* method in shown in orange. Errors in eigenfunctions for $l = 0$ to $l = 5$ with $m \in [0, l]$ are shown from left to right and top to bottom. The error value decreases with the increase in number of vertices in the mesh, and the proposed method performs better consistently.

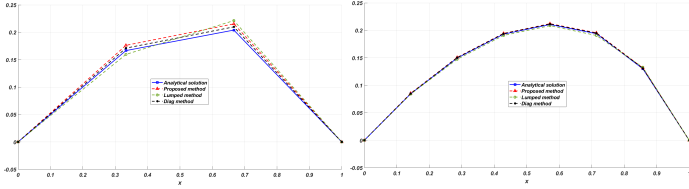


Figure 13: Analytical and Numerical solutions of the Poisson equation with right hand side function $f(x) = \sin(x) + e^x + x^3$. The solutions are shown for 4(left) and 8(right) points resp. The analytical solution, *Lumped* and *Diag* method solutions are shown in blue, red, green and black resp.

used for this experiment are: $f(x) = \sin(x) + e^x + x^3$ and $f(\theta, \phi) = 12 \sin(\theta) \cos(\psi)$ for the interval and sphere resp. For the Poisson problem, the time taken by the proposed method is lower compared to the lumped method due to the extra computations needed to compute $A_l A^{-1} \bar{f}$ for the lumped method. If we use $A_l A^{-1} \bar{f} \simeq \bar{f}$, observe that the required time for the lumped method drops below that of the proposed method as expected. It is to be noted that the computational times for the proposed and lumped method are comparable.

Table 1: Average and std. deviation of computation time (in seconds) for 100 runs of the Eigenvalue problem for different meshes

Mesh	Lumped($\hat{H}_{\text{lump}}$)	Proposed($\hat{H}_{\text{prop}}$)
Cactus (5261)	0.559 ± 0.033	0.564 ± 0.043
Centaur (15768)	1.414 ± 0.086	1.465 ± 0.166
StBunny (55684)	7.297 ± 0.574	7.894 ± 0.767
Armadillo (172974)	30.330 ± 5.085	31.156 ± 4.939

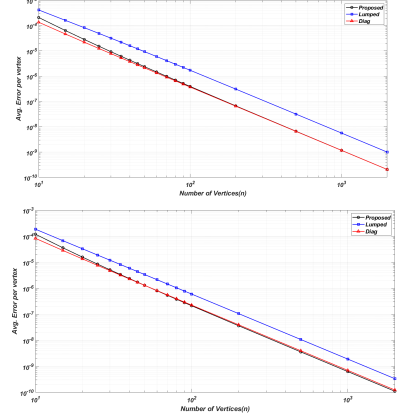


Figure 14: Error in three different numerical solutions for the right hand side function (top) $f(x) = \sin(x) + e^x + x^3$ and $V \equiv 4$, and (bottom) $f(x) = \sin(x) + e^x + x^3$ and $V(x) = \exp(x)$ on $[0, 1]$, from 5 to 2000 sampled points. Notice that the error is higher in the *Lumped* method compared to the *Diag* and proposed methods across all resolutions.

6 Conclusion

In this paper, we proposed a FEM based discretisation of the Hamiltonian operator on triangular meshes. We proved that this discretisation has several nice properties including symmetry, orthonormality of eigenvectors, non-negativity of eigenvalues. The *Lumped* matrix approximation also preserves the orthonormality of eigenvectors. But for the constant potential function in which case the eigenfunctions of the Hamiltonian operator are those of the LBO, there is larger error for the *Lumped* approximation in comparison to our scheme. We have also proved that the *Diag* and the proposed discretisations are the same if and only if the potential function is a constant.

On the unit interval as well as the unit sphere, the eigenvectors of the Hamiltonian computed through the proposed method are closer to the analytical eigenfunctions as compared to the ones computed through the *Lumped* method. For the discrete Poisson equation (with Hamiltonian operator), our method has better accuracy

Table 2: Average and std. deviation of computation time (in seconds) for 100 runs of the Poisson problem for different meshes. For the lumped approximation, computation time for the right hand side as $A_I A^{-1} \bar{f}$ and $\bar{f}$ are given. For the proposed method, $\bar{f}$ is used.

Mesh	Lumped ($\hat{H}_{\text{lump}}$)		Proposed ($\hat{H}_{\text{prop}}$)
	RHS used $\rightarrow$	$A_I A^{-1} \bar{f}$	$\bar{f}$
1D (500)		0.027 $\pm$ 0.004	0.025 $\pm$ 0.004
Sphere (642)		0.006 $\pm$ 0.002	0.003 $\pm$ 0.002
Sphere (2562)		0.027 $\pm$ 0.006	0.018 $\pm$ 0.005
Sphere (10242)		0.119 $\pm$ 0.011	0.079 $\pm$ 0.012

at lower resolution.

It would be interesting to see the implications of the non-orthogonality of eigenvectors of $\hat{H}_{\text{diag}}$ on applications. We would also like to explore similar discretisations of other operators, for instance, the Dirichlet-to-Neumann operator [14].

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A Existence and Uniqueness of the Poisson Equation

Theorem A.1. Assume that $V \in L^\infty(a, b)$ and $f \in L^2(a, b)$. If $V(x) \geq 0$ almost everywhere in (a, b) , then the problem

$$\int_a^b (u'(x)\varphi'(x) + V(x)u(x)\varphi(x)) dx = \int_a^b f(x)\varphi(x) dx, \quad \forall \varphi \in H_0^1(a, b), \quad (18)$$

admits a unique solution $u \in H_0^1(a, b)$. Moreover, there exists a constant $C > 0$ such that $\|u'\|_{L^2(a, b)} \leq C\|f\|_{L^2(a, b)}$.

Proof. Define the bilinear form $B(u, v) = \int_a^b (u'(x)v'(x) + V(x)u(x)v(x)) dx$, and the linear functional $L(\varphi) = \int_a^b f(x)\varphi(x) dx$. Then (18) can be written as $B(u, \varphi) = L(\varphi)$, $\forall \varphi \in H_0^1(a, b)$. We verify the assumptions of the Lax–Milgram theorem. *Continuity of B:* Using the Cauchy–Schwarz inequality and the boundedness of V , we obtain $|B(u, \varphi)| \leq \|u'\|_{L^2} \|\varphi'\|_{L^2} + \|V\|_{L^\infty} \|u\|_{L^2} \|\varphi\|_{L^2}$. By the Poincaré inequality, $\|u\|_{L^2} \leq C_P \|u'\|_{L^2}$, hence $|B(u, \varphi)| \leq C \|u'\|_{L^2} \|\varphi'\|_{L^2}$, so B is continuous on $H_0^1(a, b)$.

Coercivity of B: Since $V(x) \geq 0$ almost everywhere, $B(u, u) = \int_a^b ((u')^2 + V(x)u^2) dx \geq \|u'\|_{L^2(a, b)}^2$. Using again the Poincaré inequality, we deduce $B(u, u) \geq c \|u\|_{H_0^1(a, b)}^2$, so B is coercive.

Continuity of L: By Cauchy–Schwarz and Poincaré inequalities, $|L(\varphi)| \leq \|f\|_{L^2(a, b)} \|\varphi\|_{L^2(a, b)} \leq C_P \|f\|_{L^2(a, b)} \|\varphi'\|_{L^2(a, b)}$. Thus L is continuous on $H_0^1(a, b)$. The result now follows from the Lax–Milgram theorem. $\square$

B Error in Poisson equation

We provide few more examples to compare errors in the three schemes while solving the Poisson equation on the unit interval and surface of the unit sphere with constant and non-constant but continuous potential.

Example 1: For $f(x) = e^x \cos(x)$, $x \in [0, 1]$, and $V \equiv 4$, and for $f(x) = 2 + x(1 - x)\sin(\pi x)$, $x \in [0, 1]$, and $V(x) = \sin(\pi x)$, $x \in [0, 1]$, the analytical solutions are $u(x) = \left(\frac{2e^x - e(2\cos(1) - \sin(1))}{2(5e^2 - e^{-2})} \right) e^{2x} + \frac{e^x}{10} (2\cos(x) - \sin(x)) +$

$\left(-\frac{1}{5} - \frac{2e^{-2} - e(2\cos(1) - \sin(1))}{2(5e^2 - e^{-2})}\right) e^{-2x}$ and $u(x) = x(1-x), x \in [0, 1]$ resp. The error for the three schemes against the number of samples is shown in Figure 15. In this case, the *Diag* method gives the most accurate solution, while the *Lumped* method is less accurate in all cases. Finally, we show experiments that compare the

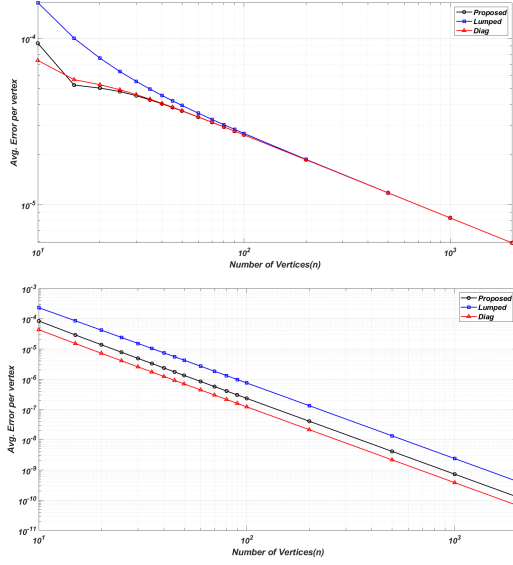


Figure 15: Error in the numerical solutions on the unit interval for (top) $f(x) = e^x \cos(x), x \in [0, 1]$ and $V \equiv 4$, and (bottom) for $f(x) = 2 + x(1-x)\sin(\pi x), x \in [0, 1]$ and $V(x) = \sin(\pi x), x \in [0, 1]$.

analytical solution of the Poisson equation on the surface of unit sphere with the numerical solutions of the three schemes discussed in this paper. The equations are written in spherical coordinates: $(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$. The Poisson equation on the sphere is given by $\left\{ -\left(\cot \theta \frac{\partial}{\partial \theta} + \frac{\partial^2}{\partial \theta^2} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) + V(\theta, \psi) \right\} u(\theta, \psi) = f(\theta, \psi)$. In these examples, the right hand side vector $\vec{f}$ as discussed in Section 5.3 is computed numerically using the MATLAB function `integral2` with Absolute error tolerance set to $1e-15$.

Example 2: For $V(\theta, \psi) \equiv 10$ and $f(\theta, \psi) = 12\sin(\theta)\cos(\psi)$, and $V(\theta, \psi) = \sin(\theta)$ and $f(\theta, \psi) = (2 + \sin(\theta))(\sin(\theta)\cos(\psi))$, the analytical solution is $u(\theta, \psi) = \sin(\theta)\cos(\psi)$ for both cases. The errors for these two cases in the three schemes are shown in Figure 16.

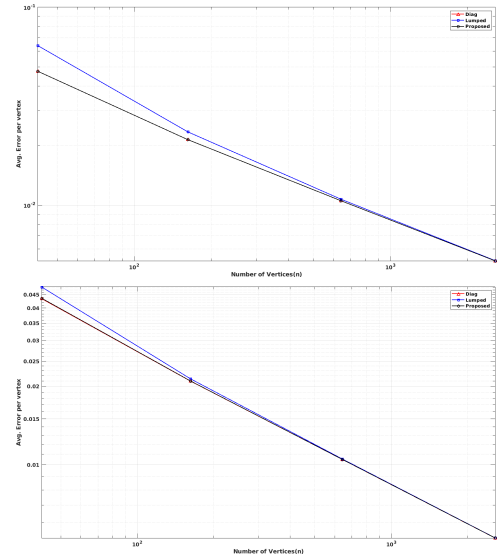


Figure 16: Error in the numerical solutions on unit sphere for (top) $f(\theta, \psi) = 12\sin(\theta)\cos(\psi)$ and $V(\theta, \psi) = 10$, and (bottom) $f(\theta, \psi) = (2 + \sin(\theta))(\sin(\theta)\cos(\psi))$ and $V(\theta, \psi) = \sin(\theta)$.